

ZETECH UNIVERSITY

RESEARCH METHODS IN I.T. (DSE 222)

FINAL RESEARCH PROPOSAL

AfyaLink: A Web-Based Healthcare Human Resource Management System (HRMS)

Prepared By

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A research proposal submitted in partial fulfilment of the requirements for the course
Research Methods in I.T. (DSE 222).

25 March 2026

DECLARATION

I, Joseph Ogwe Were, declare that this research proposal is my original work and has not been submitted to any other university, institution, or examining body for academic award. All sources used in this proposal have been acknowledged appropriately.

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APPROVAL

This research proposal has been submitted for examination with the approval of the undersigned supervisor.

Supervisor Name: _____

Signature: _____

Date: _____

DEDICATION

This proposal is dedicated to my family, mentors, and healthcare professionals whose daily commitment to patient care continues to inspire the development of better digital health systems.

ACKNOWLEDGEMENT

I acknowledge my lecturer, classmates, and the healthcare practitioners whose experiences informed this proposal. I also appreciate the support received during the analysis and design of AfyaLink.

ABSTRACT

Healthcare institutions depend on accurate staff records, secure access control, reliable attendance management, and timely reporting in order to deliver quality services. Many facilities still rely on paper-based records, spreadsheets, or fragmented digital tools that create data inconsistency, weak accountability, and delayed decision-making. This research proposes AfyaLink, a web-based Healthcare Human Resource Management System designed to improve staff management, role-based access, attendance tracking, and compliance monitoring in healthcare institutions. The study adopts a design and development research approach supported by interviews, observation, and document analysis. Agile methodology is proposed for iterative system development, while the system design is illustrated using architecture, data flow, and use case models. The expected outcome is a secure and centralized platform that improves administrative efficiency, data accuracy, transparency, and service delivery in healthcare organizations.

TABLE OF CONTENTS

DECLARATION	i
APPROVAL	ii
DEDICATION	iii
ACKNOWLEDGEMENT	iv
ABSTRACT	v
LIST OF ABBREVIATIONS	x
CHAPTER ONE: INTRODUCTION	1
1.1 Introduction	1
1.2 Background of the Study	1
1.3 Statement of the Problem	1
1.4 Objectives of the Study	1
1.4.1 General Objective	1
1.4.2 Specific Objectives	1
1.5 Research Questions	2
1.6 Justification of the Study	2
1.7 Scope of the Study	2
CHAPTER TWO: LITERATURE REVIEW	3
2.1 Introduction	3
2.2 Objectives of the Literature Review	3
2.3 Key Concepts	3
2.3.1 Human Resource Management System (HRMS)	3
2.3.2 Healthcare Information Systems	3
2.3.3 Web-Based Systems	3
2.3.4 Role-Based Access Control (RBAC)	4

2.3.5 <i>Information Security in Healthcare Systems</i>	4
2.3.6 <i>Digital Transformation in Healthcare Administration</i>	4
2.4 Summary of Literature Review	4
CHAPTER THREE: RESEARCH METHODOLOGY	5
3.1 Introduction	5
3.2 Research Design	5
3.3 Data Collection Methods	5
3.3.1 <i>Interviews</i>	5
3.3.2 <i>Observation</i>	5
3.3.3 <i>Document Analysis</i>	5
3.4 System Development Methodology	5
3.5 System Design	5
3.5.1 <i>System Architecture</i>	6
3.5.2 <i>Data Flow Diagram</i>	6
3.5.3 <i>Use Case Diagram</i>	7
3.5.4 <i>System Requirements</i>	7
3.6 System Development Tools	8
3.7 System Testing	8
3.7.1 <i>Unit Testing</i>	8
3.7.2 <i>Integration Testing</i>	8
3.7.3 <i>User Acceptance Testing</i>	8
3.8 Ethical Considerations	8
3.9 Summary	9
REFERENCES	10
APPENDICES	11
Appendix A: Proposed Budget	11
Appendix B: Questionnaire / Interview Guide	11

LIST OF FIGURES

Figure 3.1: AfyaLink system architecture.	6
Figure 3.2: AfyaLink data flow diagram.	7
Figure 3.3: AfyaLink use case diagram.	7

LIST OF TABLES

Table 3.1: Functional and non-functional requirements of AfyaLink.	7
Table 3.2: AfyaLink development tools and platform components.	8
Table 4.1: Proposed budget for the AfyaLink research proposal.	11

LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
FHIR	Fast Healthcare Interoperability Resources
HIS	Healthcare Information System
HL7	Health Level Seven
HRMS	Human Resource Management System
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface

CHAPTER ONE: INTRODUCTION

1.1 Introduction

The healthcare sector relies heavily on efficient management of human resources to deliver quality patient care. Hospitals and healthcare institutions require accurate staff records, attendance tracking, role-based access, compliance monitoring, and secure communication. However, many healthcare facilities still depend on manual or semi-digital systems that are inefficient, insecure, and error-prone. This research proposes AfyaLink, a web-based Healthcare Human Resource Management System designed to automate and streamline human resource operations in healthcare institutions.

1.2 Background of the Study

Healthcare institutions manage diverse personnel such as doctors, nurses, laboratory technicians, pharmacists, and administrative staff. Traditional HR practices like paper-based records, spreadsheets, and disconnected systems lead to challenges such as data redundancy, loss of records, delayed decision-making, and poor accountability. With advances in information technology, web-based systems offer secure, scalable, and efficient solutions. AfyaLink aims to leverage modern web technologies to provide a centralized platform for managing healthcare staff, improving efficiency, transparency, and compliance with healthcare regulations.

1.3 Statement of the Problem

Many healthcare institutions face challenges in managing human resources effectively because manual record-keeping leads to data loss and inaccuracies, access control remains weak, attendance and performance tracking are inconsistent, compliance and audit trails are difficult to monitor, and HR processes are poorly integrated with digital systems. These challenges negatively affect operational efficiency and the quality of healthcare services. There is therefore a need for a secure, reliable, and automated healthcare HR management system.

1.4 Objectives of the Study

1.4.1 General Objective

To design and propose AfyaLink, a web-based Healthcare Human Resource Management System that improves efficiency, security, and accountability in healthcare institutions.

1.4.2 Specific Objectives

1. To analyze existing HR management challenges in healthcare institutions.

2. To design a secure user authentication and role-based access system.
3. To develop a centralized staff management and record-keeping system.
4. To implement attendance monitoring, audit logging, and compliance support features.
5. To evaluate the effectiveness of the proposed system in improving HR operations.

1.5 Research Questions

1. What challenges do healthcare institutions face in managing human resources using existing systems?
2. How can a web-based HRMS improve efficiency and data accuracy in healthcare institutions?
3. What security mechanisms are required to protect sensitive healthcare staff data?
4. How effective is AfyaLink in supporting healthcare HR operations compared to traditional systems?

1.6 Justification of the Study

AfyaLink addresses the administrative and governance gaps that arise when healthcare organizations rely on fragmented tools. The proposed system brings together centralized staff records, role-based access, attendance control, reporting, and auditability in a single platform. This supports faster decision-making, stronger accountability, and improved organizational transparency.

1.7 Scope of the Study

This proposal focuses on the design of a web-based HRMS for healthcare institutions. The scope covers user authentication, role-based access control, staff record management, attendance tracking, reporting, and compliance support. The proposal does not cover detailed financial accounting, patient clinical management, or national production deployment beyond the project design stage.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to Healthcare Human Resource Management Systems, digital transformation in healthcare institutions, information security in web-based systems, and role-based access control mechanisms. The purpose of the literature review is to examine previous studies, theoretical foundations, and existing systems in order to identify gaps that justify the development of AfyaLink.

2.2 Objectives of the Literature Review

1. To examine existing Human Resource Management Systems used in healthcare institutions.
2. To explore the role of information technology in improving healthcare administrative efficiency.
3. To analyze security mechanisms used in web-based HR systems.
4. To identify gaps in current systems that justify the development of AfyaLink.

2.3 Key Concepts

2.3.1 Human Resource Management System (HRMS)

A Human Resource Management System is a digital platform designed to automate and manage human resource processes such as employee records, payroll, attendance, recruitment, performance evaluation, and compliance monitoring. HRMS platforms improve organizational efficiency by reducing manual paperwork, minimizing human error, and enabling centralized data management (Laudon & Laudon, 2022). In healthcare institutions, HRMS is important because of the complexity of staffing structures, regulatory obligations, and the need for accurate scheduling of clinical personnel.

2.3.2 Healthcare Information Systems

Healthcare Information Systems are integrated systems used to manage healthcare data, including patient information, staff records, billing, and hospital operations. Many healthcare systems focus primarily on patient management and neglect human resource management, creating administrative inefficiencies. AfyaLink seeks to bridge this gap by focusing specifically on healthcare human resource operations while still aligning with broader digital hospital workflows.

2.3.3 Web-Based Systems

A web-based system is an application accessed through a web browser over a network. Web-based systems offer remote accessibility, centralized data storage, real-time updates, scalability, and reduced installation and maintenance costs. According to Sommerville (2016), web-based information systems provide a practical architecture for organizations that need shared access and centralized control across multiple users and departments.

2.3.4 Role-Based Access Control (RBAC)

Role-Based Access Control is a security mechanism that restricts system access based on user roles. Instead of assigning permissions individually, permissions are grouped into roles such as Administrator, HR Officer, Doctor, or Nurse. Sandhu, Coyne, Feinstein, and Youman (1996) argue that RBAC enhances data security and administrative efficiency by ensuring users access only information relevant to their responsibilities. In healthcare environments where staff data is sensitive, RBAC is essential for maintaining confidentiality and compliance.

2.3.5 Information Security in Healthcare Systems

Healthcare institutions handle sensitive personal and professional data. Information security principles such as confidentiality, integrity, and availability are critical in protecting healthcare information systems. Cybersecurity threats such as unauthorized access, data breaches, and system misuse can compromise institutional trust and operational efficiency. The Kenya Data Protection Act (Government of Kenya, 2019) reinforces the need for lawful data handling, security safeguards, and responsible processing of personal data in digital systems.

2.3.6 Digital Transformation in Healthcare Administration

Digital transformation refers to the integration of digital technologies into all aspects of organizational operations. In healthcare administration, digital transformation improves decision-making speed, record accuracy, transparency, compliance monitoring, and resource allocation. Despite global trends toward digitization, many institutions still rely on manual HR processes. This creates a need for affordable, secure, and scalable digital HR solutions such as AfyaLink.

2.4 Summary of Literature Review

The reviewed literature demonstrates that HRMS improves efficiency and reduces administrative errors. Web-based systems provide scalability and accessibility advantages, while security mechanisms such as RBAC are essential in healthcare environments. However, there remains a gap in specialized, secure, and healthcare-focused HR management systems tailored to institutional needs. AfyaLink addresses this gap by providing a centralized, web-based, and secure HRMS designed specifically for healthcare institutions.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the research methodology used in the development of AfyaLink, a web-based Healthcare Human Resource Management System. It outlines the research design, data collection methods, system development methodology, system design approach, development tools, and testing techniques used in the study.

3.2 Research Design

This study adopts a design and development research approach. The approach is appropriate because it focuses on designing, developing, and evaluating a functional system to solve real-world challenges in healthcare human resource management. The process involves problem identification, requirements analysis, system design, system development, and testing.

3.3 Data Collection Methods

3.3.1 Interviews

Interviews are proposed with healthcare administrators, HR officers, and staff members in order to gather detailed information on current HR practices, pain points, reporting needs, and expectations for a digital solution.

3.3.2 Observation

Observation is used to examine how HR activities such as staff registration, attendance tracking, approvals, and report preparation are carried out in practice. This helps identify actual workflow gaps that may not be captured fully through interviews alone.

3.3.3 Document Analysis

Document analysis is used to review existing staff records, attendance logs, policy documents, and related administrative forms. The reviewed documents help identify duplication, delays, and control weaknesses in the current process.

3.4 System Development Methodology

The system is developed using Agile methodology because Agile supports iterative delivery, regular feedback, and continuous refinement during development (Pressman & Maxim, 2020). This is suitable for AfyaLink because system requirements can be validated progressively with intended users during the design and prototyping stages.

3.5 System Design

3.5.1 System Architecture

AfyaLink uses a client-server architecture in which users access the system through web browsers, business logic is processed by a Node.js and Express backend, and data is stored in MongoDB and Redis. The project also supports observability and integration services that improve reliability and extensibility.

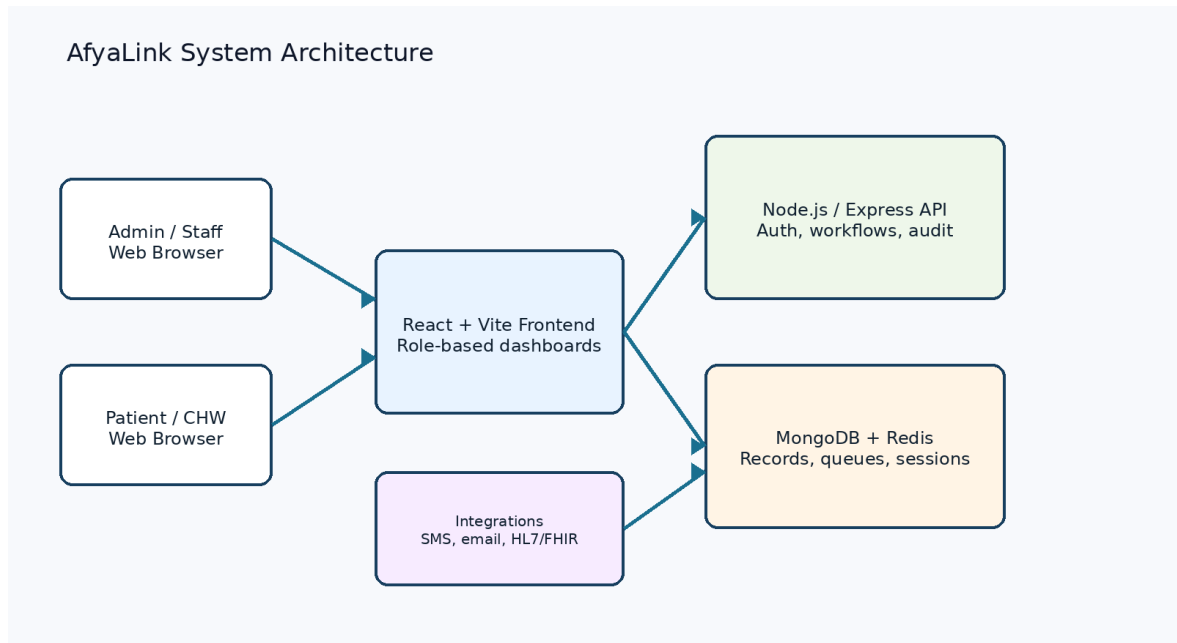


Figure 3.1: AfyaLink system architecture.

3.5.2 Data Flow Diagram

The data flow diagram shows how users submit requests to the AfyaLink system, how the system processes those requests, and how data is stored and retrieved from the database to support staff administration, attendance management, and reporting.

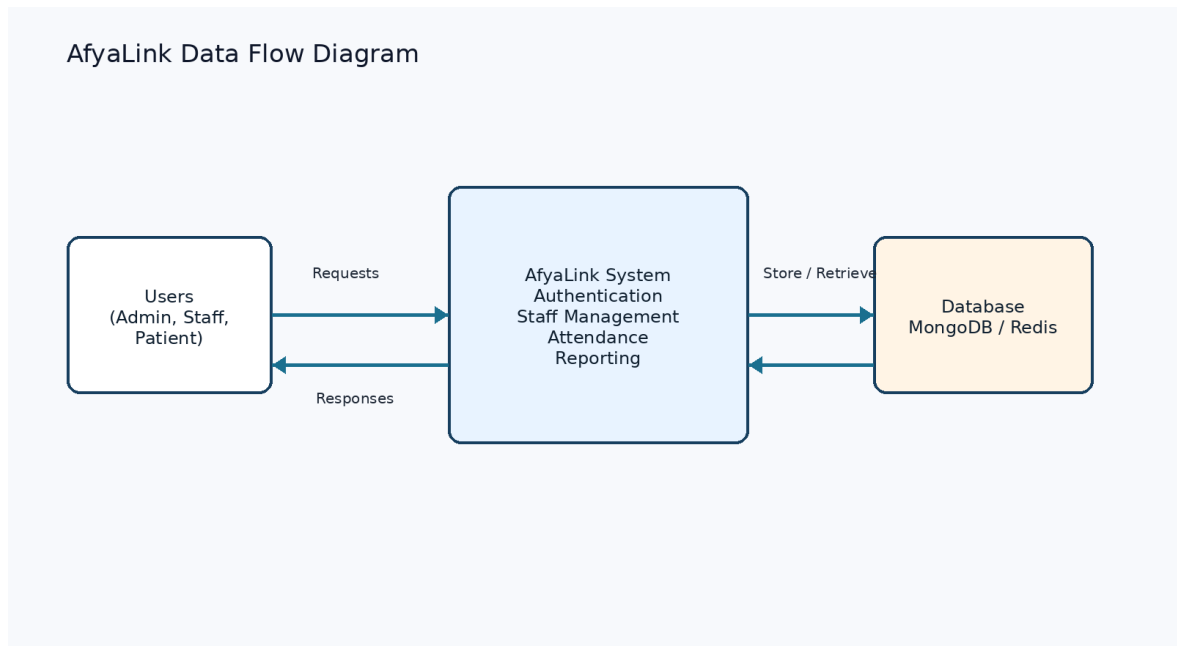


Figure 3.2: AfyaLink data flow diagram.

3.5.3 Use Case Diagram

The use case diagram summarizes the main interactions between users and the system. Administrative users are expected to log in, manage staff details, track attendance, and generate reports, while other staff interact with the relevant modules based on their assigned roles.

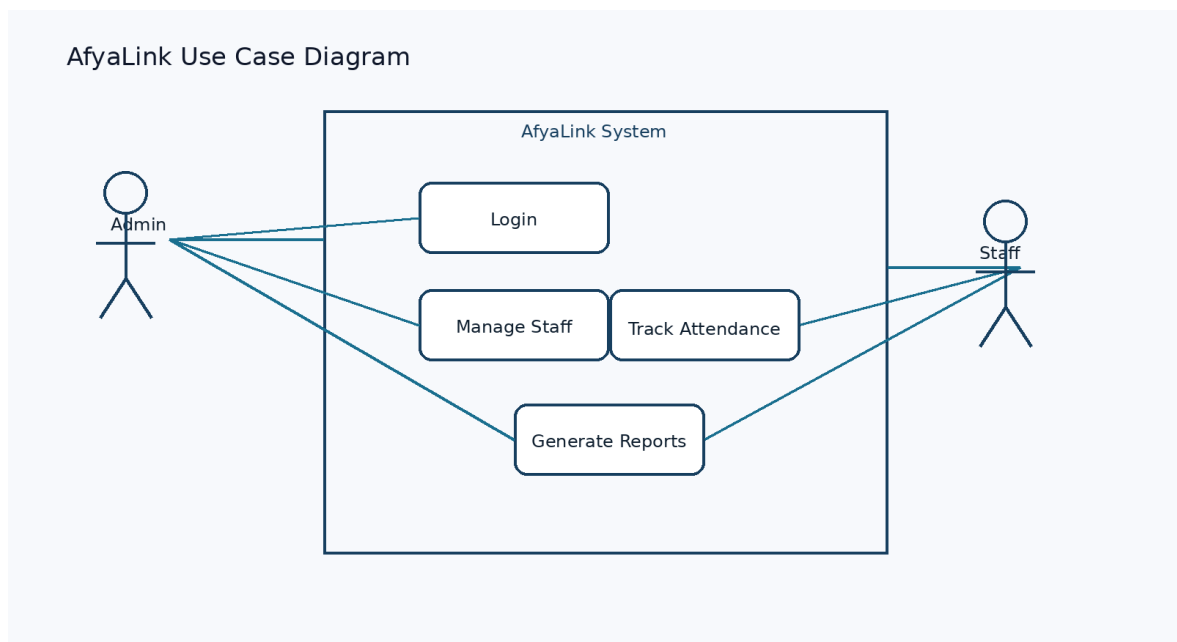


Figure 3.3: AfyaLink use case diagram.

3.5.4 System Requirements

Table 3.1: Functional and non-functional requirements of AfyaLink.

Category	Requirements
Functional Requirements	User authentication, role-based access control, staff management, attendance tracking, and reporting
Non-Functional Requirements	Security, scalability, usability, reliability, and data integrity.

3.6 System Development Tools

Table 3.2: AfyaLink development tools and platform components.

Component	Tools / Technologies
Frontend	React, Vite, React Router, LocalForage
Backend	Node.js, Express, Socket.IO
Database	MongoDB, Redis
Integration Support	HL7, FHIR tooling, SendGrid, Africa's Talking, Twilio
Version Control	Git
Reliability / Observability	Prometheus, Grafana, alert rules, readiness gates

3.7 System Testing

3.7.1 Unit Testing

Individual components are tested independently to confirm that each module performs its expected function correctly.

3.7.2 Integration Testing

Modules are tested together to ensure that data passes correctly between authentication, staff management, attendance, and reporting workflows.

3.7.3 User Acceptance Testing

Intended users review the prototype to confirm that the system meets operational requirements and is usable in real administrative settings.

3.8 Ethical Considerations

Ethical standards are observed throughout the proposed study. Information obtained from respondents is treated confidentially and used strictly for academic purposes. The proposal also recognizes the need to comply with the Kenya Data Protection Act, 2019 when handling personal data collected during system analysis and future deployment.

3.9 Summary

This chapter has described the methodology used in developing AfyaLink, including the research design, data collection methods, system design approach, development tools, and testing techniques. The methodology provides a structured foundation for implementing the proposed system.

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APPENDICES

Appendix A: Proposed Budget

Table 4.1: Proposed budget for the AfyaLink research proposal.

Item	Quantity	Unit Cost (KES)	Total (KES)
Printing and stationery	1	3,000	3,000
Internet and data bundles	3 months	2,500	7,500
Transport for data collection	6 visits	2,000	12,000
Airtime and communication	3 months	1,500	4,500
Laptop and power maintenance	1	5,000	5,000
Binding and submission	2 copies	1,000	2,000
Contingency	1	6,000	6,000
Total			40,000

Appendix B: Questionnaire / Interview Guide

Section A: Respondent Profile

1. What is your role in the healthcare institution?
2. How long have you worked in your current position?

Section B: Current HR Practices

3. How are staff records currently created, stored, and updated in your institution?
4. What challenges do you experience when managing staff attendance and reporting?
5. What delays or errors are common in the current HR process?

Section C: Security and Access Control

6. Who is allowed to access staff records, and how is access currently controlled?
7. What security concerns do you have about the current process?

Section D: Expectations for AfyaLink

8. Which HR activities should be prioritized in a digital system for your institution?
9. What reports or dashboards would help management make better decisions?
10. What features would make a new HR system easier for staff to adopt?